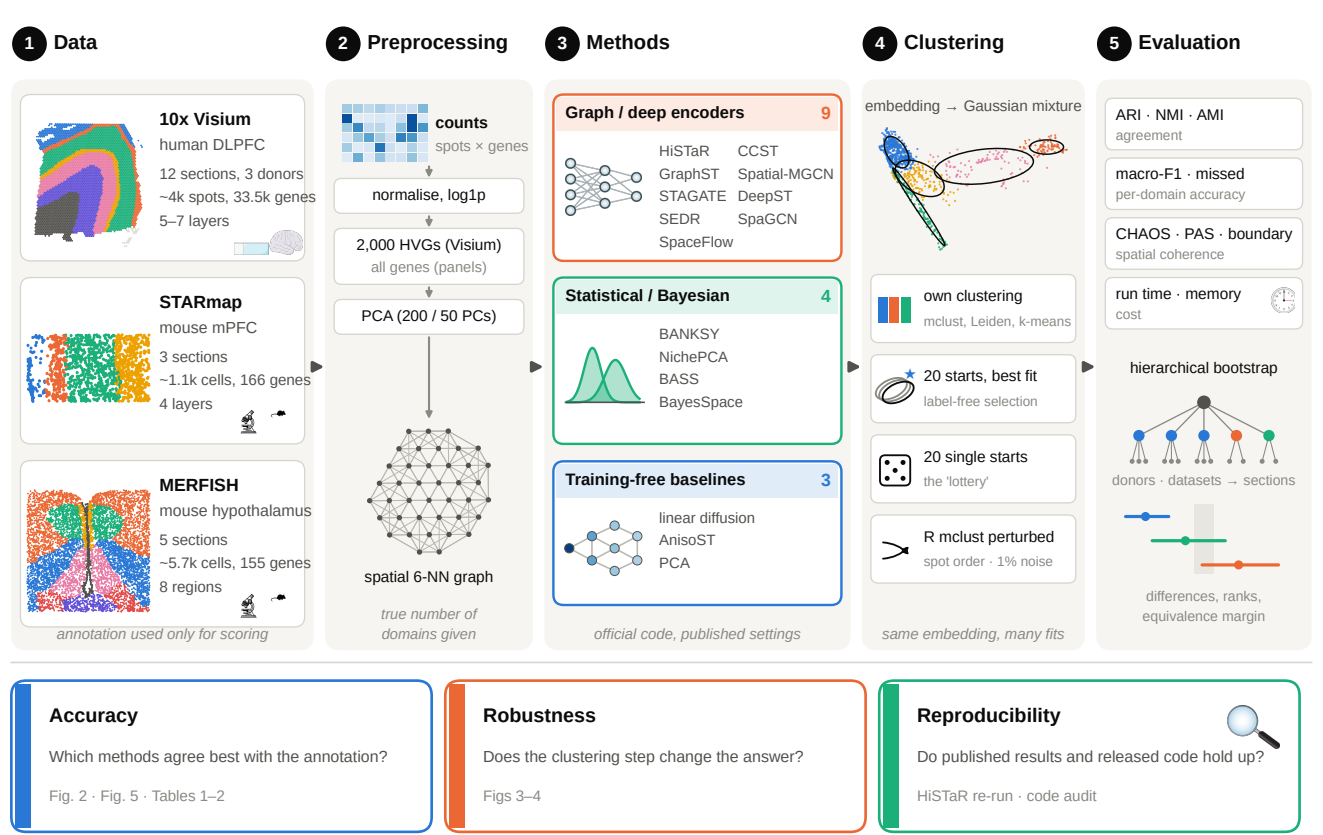

1 Data
10x Visium
human DLPFC
12 sections, 3 donors
~4k spots, 33.5k genes
5–7 layers
STARmap
mouse mPFC
3 sections
~1.1k cells, 166 genes
4 layers
MERFISH
mouse hypothalamus
5 sections
~5.7k cells, 155 genes
8 regions
annotation used only for scoring
2 Preprocessing
counts
spots × genes
normalise, log1p
2,000 HVGs (Visium)
all genes (panels)
PCA (200 / 50 PCs)
spatial 6-NN graph
true number of domains given
3 Methods
Graph / deep encoders 9
HiSTaR
CCST
GraphST
Spatial-MGCN
STAGATE
DeepST
SEDR
SpaGCN
SpaceFlow
Statistical / Bayesian 4
BANKSY
NichePCA
BASS
BayesSpace
Training-free baselines 3
linear diffusion
AnisoST
PCA
official code, published settings
4 Clustering
embedding → Gaussian mixture
own clustering
mclust, Leiden, k-means
20 starts, best fit
label-free selection
20 single starts
the 'lottery'
R mclust perturbed
spot order · 1% noise
same embedding, many fits
5 Evaluation
ARI · NMI · AMI
agreement
macro-F1 · missed
per-domain accuracy
CHAOS · PAS · boundary
spatial coherence
run time · memory
cost
hierarchical bootstrap
donors · datasets → sections
differences, ranks,
equivalence margin
Accuracy
Which methods agree best with the annotation?
Fig. 2 · Fig. 5 · Tables 1–2
Robustness
Does the clustering step change the answer?
Figs 3–4
Reproducibility
Do published results and released code hold up?
HiSTaR re-run · code audit